```
clc;
clear;
close all;

%% PARAMETERS
A = 1;
f = 50;
Fs = 1000;
N = 1000;

n = 0:N-1;
x = A*sin(2*pi*f*n/Fs);

B = 8;
bits = [4 8 12 16];

%% 8-BIT QUANTIZATION
D = 2*A/2^B;

xr = D*round(x/D);       % Rounding
xt = D*fix(x/D);         % Truncation

er = x - xr;             % Rounding error
et = x - xt;             % Truncation error

SQr = 10*log10(mean(x.^2)/mean(er.^2));
SQt = 10*log10(mean(x.^2)/mean(et.^2));

%% INPUT / OUTPUT QUANTIZATION
y = x;
yq = D*round(y/D);
eo = y - yq;

SQout = 10*log10(mean(y.^2)/mean(eo.^2));

%% WORD-LENGTH COMPARISON
step = zeros(1,length(bits));
sqnrR = zeros(1,length(bits));
sqnrT = zeros(1,length(bits));
maxER = zeros(1,length(bits));
maxET = zeros(1,length(bits));

for k = 1:length(bits)
    b = bits(k);
    d = 2*A/2^b;

    qr = d*round(x/d);
    qt = d*fix(x/d);

    er1 = x - qr;
```

```
    et1 = x - qt;

    step(k) = d;
    sqnrR(k) = 10*log10(mean(x.^2)/mean(er1.^2));
    sqnrT(k) = 10*log10(mean(x.^2)/mean(et1.^2));

    maxER(k) = max(abs(er1));
    maxET(k) = max(abs(et1));
end

%% DISPLAY RESULTS
fprintf('\n========== QUANTIZATION RESULTS ==========\n');
```

```
========== QUANTIZATION RESULTS ==========
```

```
fprintf('Amplitude = %.1f | Frequency = %d Hz | Fs = %d Hz\n', A, f, Fs);
```

```
Amplitude = 1.0 | Frequency = 50 Hz | Fs = 1000 Hz
```

```
fprintf('Samples = %d | Main word length = %d bits\n', N, B);
```

```
Samples = 1000 | Main word length = 8 bits
```

```
fprintf('8-bit Step Size = %.7f\n', D);
```

```
8-bit Step Size = 0.0078125
```

```
fprintf('Rounding SQNR = %.4f dB\n', SQr);
```

```
Rounding SQNR = 48.9339 dB
```

```
fprintf('Truncation SQNR = %.4f dB\n', SQt);
```

```
Truncation SQNR = 45.2931 dB
```

```
fprintf('Output SQNR = %.4f dB\n', SQout);
```

```
Output SQNR = 48.9339 dB
```

```
fprintf('Max Rounding Error = %.6f\n', max(abs(er)));
```

```
Max Rounding Error = 0.003483
```

```
fprintf('Max Truncation Error = %.6f\n', max(abs(et)));
```

```
Max Truncation Error = 0.005744
```

```

%% FIRST 10 SAMPLES
T1 = table(n(1:10)', x(1:10)', xr(1:10)', ...
    er(1:10)', xt(1:10)', et(1:10)', ...
    'VariableNames', {'n','Original','Rounded', ...
    'RoundError','Truncated','TruncError'});

disp(T1);
```

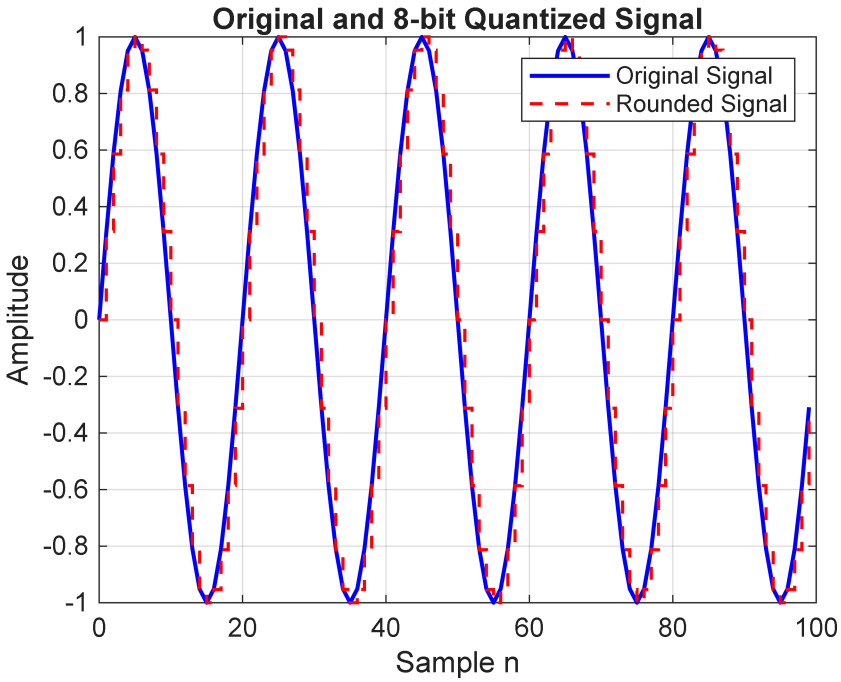
n	Original	Rounded	RoundError	Truncated	TruncError
0	0	0	0	0	0
1	0.30902	0.3125	-0.003483	0.30469	0.0043295
2	0.58779	0.58594	0.0018478	0.58594	0.0018478
3	0.80902	0.8125	-0.003483	0.80469	0.0043295
4	0.95106	0.95312	-0.0020685	0.94531	0.005744
5	1	1	0	1	0
6	0.95106	0.95312	-0.0020685	0.94531	0.005744
7	0.80902	0.8125	-0.003483	0.80469	0.0043295
8	0.58779	0.58594	0.0018478	0.58594	0.0018478
9	0.30902	0.3125	-0.003483	0.30469	0.0043295

```
%% WORD-LENGTH TABLE
T2 = table(bits', step', sqnrR', sqnrT', maxER', maxET', ...
    'VariableNames', {'Bits','StepSize','SQNR_Round_dB', ...
    'SQNR_Trunc_dB','MaxError_Round','MaxError_Trunc'});

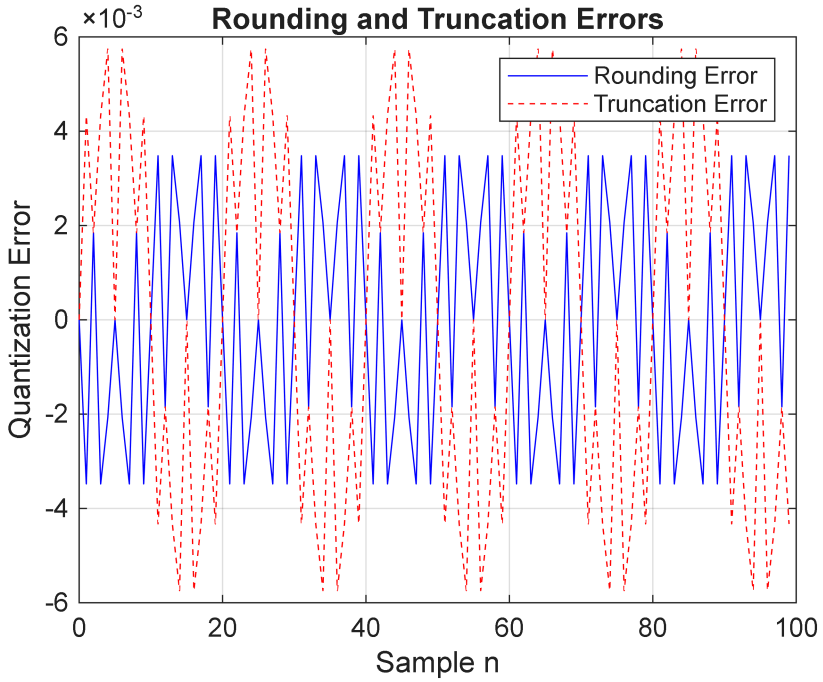
disp(T2);
```

Bits	StepSize	SQNR_Round_dB	SQNR_Trunc_dB	MaxError_Round	MaxError_Trunc
4	0.125	23.667	20.871	0.059017	0.087785
8	0.0078125	48.934	45.293	0.003483	0.005744
12	0.00048828	78.811	65.891	0.00011536	0.00042324
16	3.0518e-05	99.697	91.599	1.382e-05	2.6516e-05

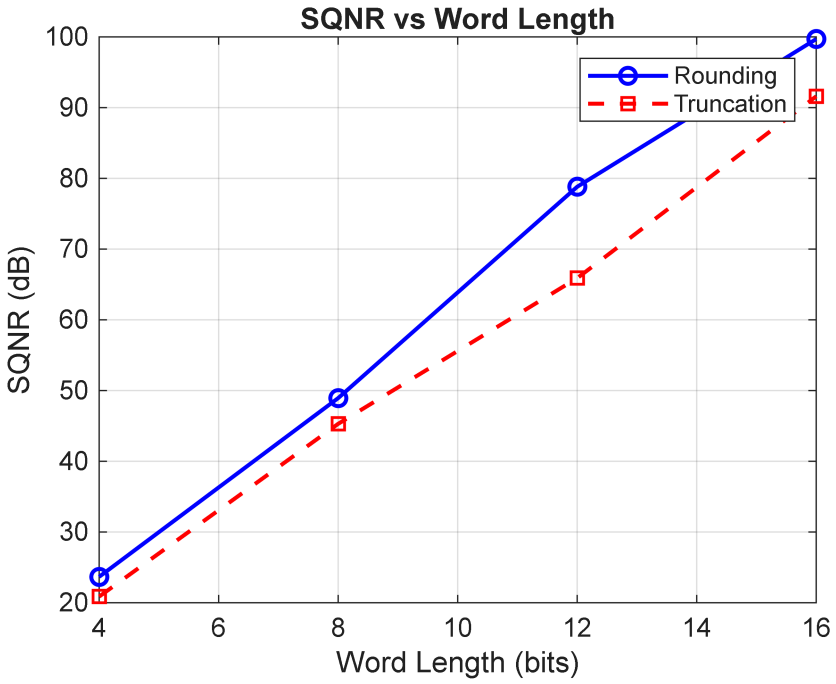
```
%% GRAPH 1: ORIGINAL AND QUANTIZED SIGNALS
figure;
plot(n(1:100), x(1:100), 'b', 'LineWidth', 1.5);
hold on;
stairs(n(1:100), xr(1:100), 'r--', 'LineWidth', 1.2);
grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Original and 8-bit Quantized Signal');
legend('Original Signal','Rounded Signal');
```




```
%% GRAPH 2: QUANTIZATION ERRORS
figure;
plot(n(1:100), er(1:100), 'b');
hold on;
plot(n(1:100), et(1:100), 'r--');
grid on;
xlabel('Sample n');
ylabel('Quantization Error');
title('Rounding and Truncation Errors');
legend('Rounding Error','Truncation Error');
```




```
%% GRAPH 3: SQNR VS WORD LENGTH
figure;
plot(bits, sqnrR, 'bo-', 'LineWidth', 1.5);
hold on;
plot(bits, sqnrT, 'rs--', 'LineWidth', 1.5);
grid on;
xlabel('Word Length (bits)');
ylabel('SQNR (dB)');
title('SQNR vs Word Length');
legend('Rounding','Truncation');
```




```
%% FINAL RESULT
fprintf('\n========== FINAL RESULT ==========\n');
```

```
========== FINAL RESULT ==========
```

```
fprintf('Uniform quantization completed successfully.\n');
```

```
Uniform quantization completed successfully.
```

```
fprintf('Quantization error and SQNR were calculated.\n');
```

```
Quantization error and SQNR were calculated.
```

```
fprintf('SQNR generally increases as word length increases.\n');
```

```
SQNR generally increases as word length increases.
```